

QTM 220

Observational Studies with Measured Confounding

Outline

Choosing Conditioning Variables

Estimation with Regression

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Omitted Variable Bias

Suppose we believe that a two variable linear model is the true model in the Washington (2008) example,

$$Y_i = \beta_0 + \beta_1 \text{Girls}_i + \beta_2 \text{Children}_i + \epsilon_i$$

but suppose that we omit the *Children* variable from the analysis

$$Y_i = \beta_0^* + \beta_1^* \text{Girls}_i + \epsilon_i^*$$

If we write the model for children as the following:

$$\textit{Children}_i = \gamma_0 + \gamma_1 \textit{Girls}_i + \nu_i$$

then the two explanatory variable model can be re-written as the following:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 \textit{Girls}_i + \beta_2 \textit{Children}_i + \epsilon_i \\ &= \beta_0 + \beta_1 \textit{Girls}_i + \beta_2 (\gamma_0 + \gamma_1 \textit{Girls}_i + \nu_i) + \epsilon_i \\ &= \underbrace{\beta_0 + \beta_2 \gamma_0}_{\beta_0^*} + \underbrace{(\beta_1 + \beta_2 \gamma_1)}_{\beta_1^*} \textit{Girls}_i + \underbrace{\epsilon_i + \beta_2 \nu_i}_{\epsilon_i^*} \end{aligned}$$

Points of Potential Confusion

Is β_1^* meant to be causal?

$$Y_i = \beta_0^* + \beta_1^* \textit{Girls}_i + \epsilon_i^*$$

Is γ_1 meant to be causal?

$$\textit{Children}_i = \gamma_0 + \gamma_1 \textit{Girls}_i + \nu_i$$

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Some Typical Econometric Advice

In sum, if the variable we leave out is positively correlated with the variable we include and also has itself a positive effect on the dependent variable, then our misspecified model will produce an upward-biased estimator of the effect of X_1 on Y . [Ashenfelter, Levine, and Zimmerman, p. 189]

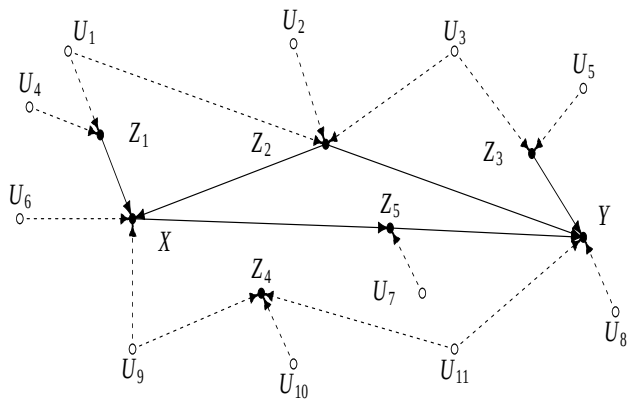
The short answer to the questions posed above is that the OLS estimates based on the equation containing too many explanatory variables will indeed be unbiased estimates of the true parameters. [Ashenfelter, Levine, and Zimmerman, p. 187]

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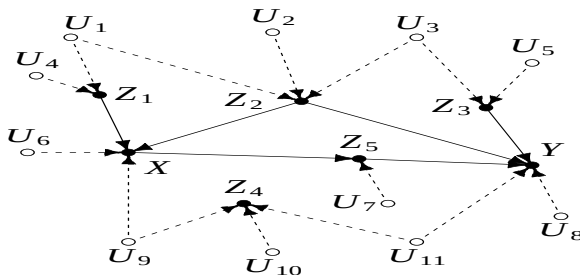
Directed Acyclic Graphs



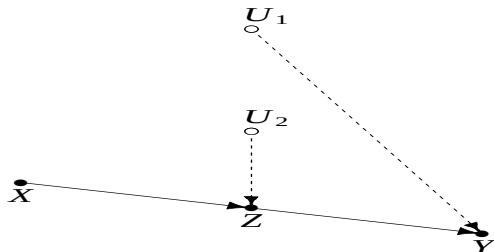
Choosing Conditioning Variables

Definition (Back-Door Criterion (Pearl (2000, p. 79)))

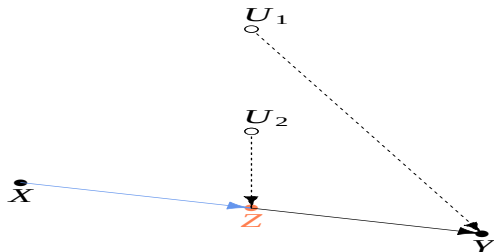
1. none of the conditioning variables is post- X ; and
2. X is “ d -separated” from Y by the conditioning variables in the graph formed by deleting all arrows out of X .



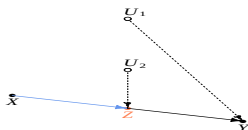
BD1: Don't condition on post-treatment variables



BD1: Don't condition on descendants of X



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In sum, if the variable we leave out is positively correlated with the variable we include and also has itself a positive effect on the dependent variable, then our misspecified model will produce an upward-biased estimator of the effect of X_1 on Y . [Ashenfelter, Levine, and Zimmerman, p. 2003]

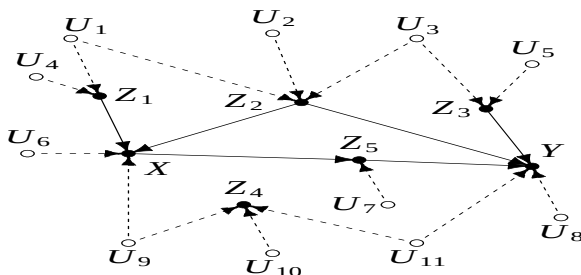
BD2: X is d -separated from Y by the conditioning variables in the graph formed by deleting all arrows out of X .

Definition (d -Separation (Pearl (2000, p. 16)))

1. X is d -Separated from Y if all paths are blocked
2. paths are blocked by “unconditioned colliders” or conditioned non-colliders

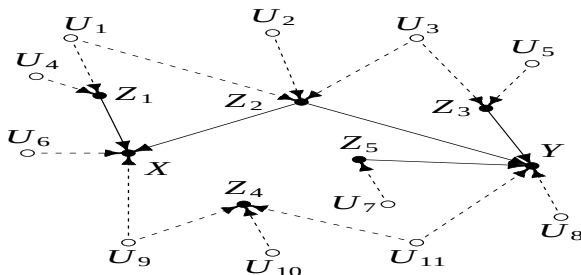
d -Separation allows conditional independence relations to be read from a DAG consistent with a probabilistic model.

Example: Sufficient Conditioning Sets



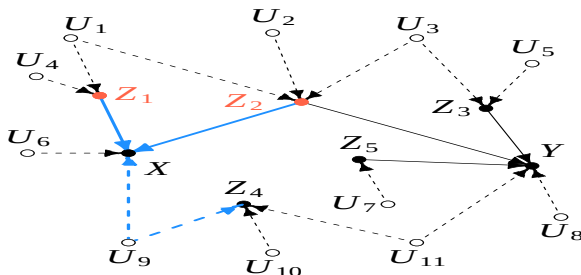
Remove arrows out of X .

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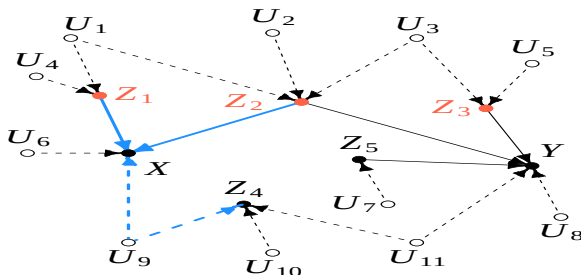
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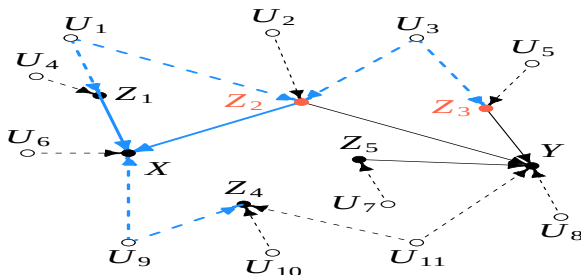
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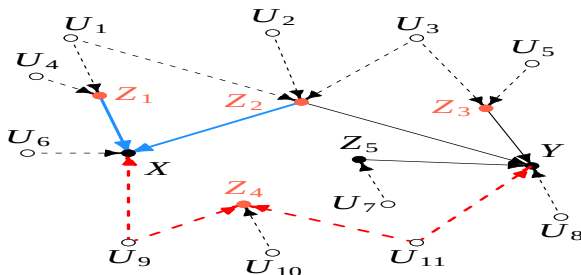
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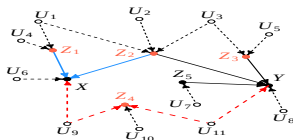
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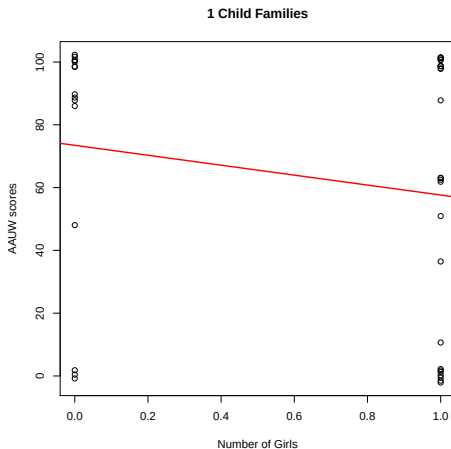
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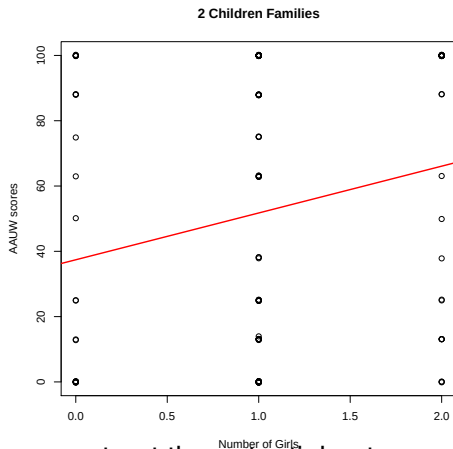
Estimation with Regression

One Child Legislators in the Washington (2008) Data



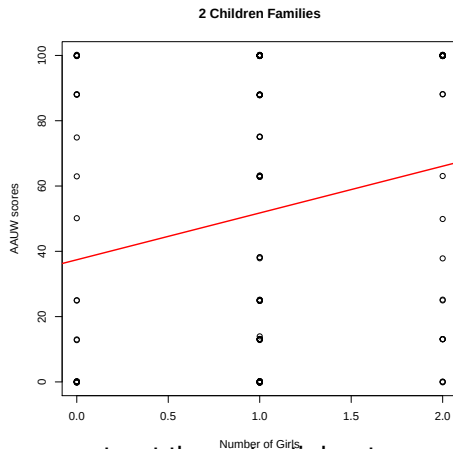
Among the subset of one-child legislators, we could adopt the standard potential outcomes approach.

Two Children Legislators in the Washington (2008) Data



How would you construct the potential outcomes table for families with 2 children? What assumption is now implicit in the regression?

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How would you construct the potential outcomes table for families with 2 children? What assumption is now implicit in the regression?

Among the subset of 2 children legislators, assume a linear CEF

$$E[Y|T = t] = \beta_0 + \beta_1 \cdot t$$

When is β_1 interpretable as the ATE of a one unit increase?

$$\begin{aligned}\beta_1 &= E[Y|T = t + 1] - E[Y|T = t] \text{ for all } t \\ &= E[Y(t+1)|T = t + 1] - E[Y(t)|T = t] \text{ for all } t \\ &= E[Y(t+1) - Y(t)] \text{ for all } t\end{aligned}$$

This linear approach would also be straightforward for 3 children, 4 children, etc.

What are some ways we could combine these analyses across all legislators?

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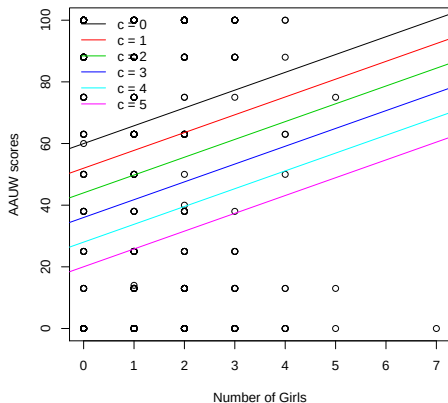
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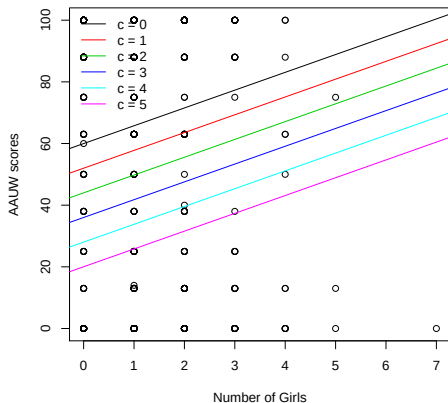
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The Additive Linear Model



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The Interactive Linear Model

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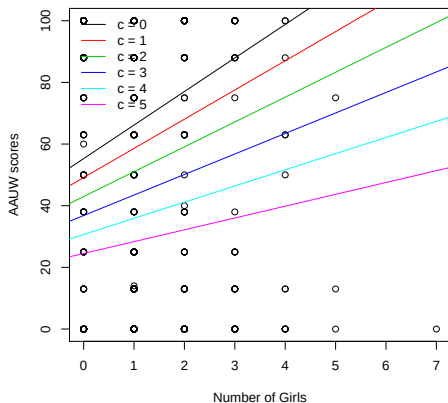
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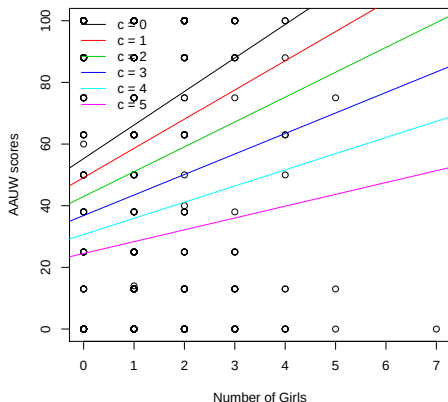
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When is $\beta_1 + \beta_3 \cdot E_X\{X\}$ interpretable as the ATE of a one unit increase?

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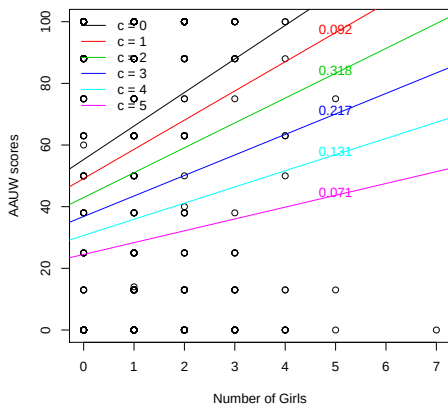
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Summary

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